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# Lab 4: VibeCoding -- Chat History
## Hangman Challenge (Scenario 16)
### Tool Used: Antigravity IDE (Claude Opus 4.6 Thinking)
### Date: October 1, 2026
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Task Overview

Repository: https://github.com/SETAPESU26/16_hangman

Objective: Fix broken code and add features to a Python-based Hangman game using vibe coding (AI-assisted)

Prompt 1 -- Understanding the Project

Me: This is my project link: https://github.com/SETAPESU26/16_hangman. Please follow exactly what instructions are given.

AI Response:

- Read the lab handout PDF and the repository README
- Cloned the repository to `C:\Users\darsh\Downloads\SE-Lab-4\16\_hangman`
- Identified 4 tasks from the README
- Identified the files: `main.py`, `game.py`, `words.py`, `stats.py`, `requirements.txt`

Task 1 -- Fix Guess-State Correctness (Bug Fix)

Bug Found:

In `game.py`, the `guess()` method only checked `self.guessed` (correct letters) for duplicate detection.

Before (Broken):

```
```python
if letter in self.guessed:
 return "Already guessed."
```
```

After (Fixed):

```
```python
if letter in self.guessed or letter in self.wrong:
 return "Already guessed."
```
```

Verification:

```

First guess z: Wrong., lives=5

Second guess z: Already guessed., lives=5 <- Fixed! Lives stayed at 5

```

Commit: `Task 1: Fix repeated wrong guess consuming extra life`

Task 2 -- Complete the Session Model

Prompt: Integrate the supplied `SessionStats` class so rounds played, rounds won, and best streak are tracked.

Changes Made to `game.py`:

1. Added `from stats import SessionStats` import
2. Created `self.stats = SessionStats()` in `\_\_init\_\_`
3. Called `self.stats.record(True, self.streak)` after winning a round
4. Called `self.stats.record(False, self.streak)` after losing a round
5. Enhanced end-of-session summary to display session statistics

```
**Verification:**  
```
```

```
After round 1 win: rounds=1, wins=1, streak=1, best=1
After round 2 loss: rounds=2, wins=1, streak=0, best=1
After round 3 win: rounds=3, wins=2, streak=1, best=1
Task 2 PASS
```
```

```
**Commit:** `Task 2: Integrate SessionStats for session-level tracking`
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```

```
## Task 3 -- Difficulty and Scoring
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```
**Prompt:** Add difficulty choices that alter available lives and scoring. Preserve category selection and
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```
**Changes Made to `game.py`:**
```

1. Added `DIFFICULTIES` dictionary:
 - Easy: 8 lives, 1.0x score multiplier, 1pt hint penalty
 - Medium: 6 lives, 1.5x score multiplier, 2pt hint penalty
 - Hard: 4 lives, 2.0x score multiplier, 3pt hint penalty
2. Added `self.difficulty = "medium"` to `\_\_init\_\_`
3. Updated `start\_round()` to use difficulty-based lives
4. Updated `use\_hint()` to use difficulty-based penalty
5. Updated win scoring: `earned = int((5 + streak) \* multiplier)`
6. Added difficulty selection prompt in `run()`

```
**Verification:**  
```
```

```
Easy: lives=8 (expect 8) [OK]
Hard: lives=4 (expect 4) [OK]
Hard scoring: base=6, mult=2.0, earned=12 (expect 12) [OK]
Hard hint penalty: score=7 (expect 7, penalty=3) [OK]
Easy hint penalty: score=9 (expect 9, penalty=1) [OK]
```
```

```
**Commit:** `Task 3: Add difficulty system with scaled lives and scoring`
```

```
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```

```
## Task 4 -- Robust Input and Feedback
```

```
**Prompt:** Improve command handling for invalid letters, repeated commands, category selection, and hint
```

```
**Changes Made to `game.py` (play_round method):**
```

1. Empty input -> silently re-prompts
2. Unknown `/commands` -> shows `"Unknown command. Use /hint or /quit."`
3. Multi-character input -> `"Please enter a single letter."`
4. Non-alphabetic characters -> `"Only letters a-z are accepted."`
5. All invalid inputs leave game state completely unchanged
6. Slash commands grouped and handled before letter validation
7. Difficulty shown during gameplay

```
**Verification:**  
```
```

```
Number input: Enter one letter., lives_changed=False [OK]
Multi-char input: Enter one letter., lives_changed=False [OK]
Empty input: Enter one letter., lives_changed=False [OK]
Repeat wrong guess: Already guessed., lives=5 [OK]
```
```

```
**Commit:** `Task 4: Robust input handling and clear feedback`
```

```
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```

Git Commit History (4 Separate Commits)

...

a3ab6f6 Task 4: Robust input handling and clear feedback
e5d5396 Task 3: Add difficulty system with scaled lives and scoring
f68d908 Task 2: Integrate SessionStats for session-level tracking
d5d9e70 Task 1: Fix repeated wrong guess consuming extra life
89f1f5c Updated scenario title (original)
b5d8237 Initial commit (original)

...

Video Evidence

BEFORE Video (Bug Demonstration)

...

Categories: technology, science, culture
Choose category or q: technology

Word: _ _ _ _ _
Wrong: -
Lives: 6 Score: 0 Streak: 0
Letter, /hint, or /quit: z
Wrong.

Word: _ _ _ _ _
Wrong: z
Lives: 5 Score: 0 Streak: 0 <- First z: 6->5
Letter, /hint, or /quit: z
Wrong.

Word: _ _ _ _ _
Wrong: z
Lives: 4 Score: 0 Streak: 0 <- Second z: 5->4 (BUG! Same letter costs another life)
Letter, /hint, or /quit: /quit
...

AFTER Video (Bug Fixed + New Features)

...

Categories: technology, science, culture
Choose category or q: technology
Difficulties: easy, medium, hard <- NEW: Difficulty selection
Choose difficulty [easy/medium/hard]: hard

Word: _ _ _ _ _
Wrong: -
Lives: 4 Score: 0 Streak: 0 <- Hard mode: 4 lives
Difficulty: hard
Letter, /hint, or /quit: z
Wrong.

Word: _ _ _ _ _
Wrong: z
Lives: 3 Score: 0 Streak: 0
Difficulty: hard
Letter, /hint, or /quit: z
Already guessed. <- FIXED! No life lost

Word: _ _ _ _ _
Wrong: z
Lives: 3 Score: 0 Streak: 0 <- Lives stayed at 3 [OK]
Difficulty: hard

Letter, /hint, or /quit: /quit
```

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## ## Final File Structure

```

```
16_hangman/  
|-- README.md           (unchanged)  
|-- requirements.txt    (unchanged)  
|-- main.py             (unchanged)  
|-- game.py             (modified -- all 4 tasks)  
|-- words.py            (unchanged)  
\-- stats.py            (unchanged -- used as-is)  
```
```

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## ## Submission Checklist

- [x] Task 1 completed -- original defect reproduced and fixed
- [x] Tasks 2-4 completed and tested
- [x] Boundary and invalid-input cases tested
- [x] No unnecessary external dependencies added
- [x] No persistent storage added
- [x] Code remains understandable and modular
- [x] Complete LLM chat history included
- [x] 4 separate git commits (one per task)
- [x] Before video recorded
- [x] After video recorded